

Linux and SSH Challenge

Submit a single document containing the five screenshots.

Timestamp requirement: A timestamp is required for all submissions. Ensure the output of the date command from your server or local terminal, or your computer's system clock, is clearly visible within every screenshot.

Screenshot 1: Steganography

Task: Use the provided challenge image. Use steghide to embed the secret text "challenge complete". The password must be ssh-is-secure.

Screenshot: Your terminal showing the complete and successful steghide embed command and the output of the date command.

Screenshot 2: SSH Key Login

Task: Your username for the challenge server must be a favorite musician or band. Configure PuTTY to log into the server using your private key. The key comment must be First Name's CHALLENGE key.

Screenshot: Your successful PuTTY login. The terminal must show you are logged in as your musician or band account, authenticated with your key, and the output of the date command run immediately after login.

Screenshot 3: X11 Forwarding

Task: Using the same SSH session from Screenshot 2, first run the date command, and then run the xeyes command.

Screenshot: Your desktop showing the graphical xeyes window. The terminal used to launch it must be visible in the background, showing the output of the date command before the xeyes command was executed.

Screenshot 4: Key and Directory Permissions

Task: Logged into the server, run the commands needed to display the file details for two things: 1. The server's host keys. 2. Your user's authentication key file.

Screenshot: A single terminal screenshot showing the commands, their output, and the output of the date command.

Screenshot 5: Restricting User Logins

This screenshot proves you can harden the server by explicitly defining who is allowed to log in.

Task: 1. On the server, edit the main SSH server configuration file as root: `sudo nano /etc/ssh/sshd_config` 2. Add a new line at the very bottom of the file: `AllowUsers your-musician-username` 3. Save the file and exit the editor. 4. Restart the SSH service to apply the change: `sudo systemctl restart sshd` 5. Open a new PuTTY session and attempt to log in as your first name, the one used in your key comment.

Screenshot: Your screenshot must show two things at the same time: 1. The PuTTY fatal error or permission denied pop-up from the failed login attempt with your first name. 2. A separate, working PuTTY window logged in as your musician or band account clearly displaying the output of the date command.